Reproducibility Budgets for Open-Weight LLM Evaluation: A Generalizability Theory Study of 7–9B Instruction-Tuned Models

Anonymous ACL submission

Abstract

Large Language Model (LLM) evaluation results vary across reproductions, but variance sources interact, and their relative importance shifts with the benchmark format. We apply Generalizability theory (G-theory) in a fully crossed design with up to seven facets across eight open-weight instruction-tuned models (7–9B parameters) and five benchmarks, totaling approximately 2.3 million evaluation records. D-study projections show that, under binary correctness scoring, a recommended 3-prompt $\times$ 3-seed design achieves $G \geq 0.80$ with 178 items (1,602 evaluations per model), a 3.1$\times$ item reduction over single-condition designs (552 items, 552 evaluations)—trading approximately 3$\times$ more total evaluations for 3$\times$ fewer items, a tradeoff that favors multi-facet designs when item construction is costly. Model$\times$item interaction (up to 36%) indicates that models differ primarily in which items they find difficult, not in overall ability. Preliminary validation across a 3B–72B parameter range and a cross-family check (Llama-3.1-70B) confirms stable variance structure across architecture families. We release an open-source G-theory pipeline and D-study budget calculator for evaluation design optimization.

1 Introduction

In our analysis of five benchmarks, item selection drives 34–38% of evaluation variance on multiple-choice tasks while prompt wording is negligible (<3%). On the free-form reasoning task GSM8K, the structure shifts: model$\times$prompt interaction alone accounts for approximately 9%, while item selection drops to 17%. These numbers, drawn from a factorial analysis of approximately 2.3 million evaluation records, reveal that the variance structure of LLM evaluation is not a fixed property of the pipeline but shifts qualitatively with the response format. Current evaluation practice treats variance sources as independent nuisance factors and controls them one at a time: one study perturbs prompts, another varies precision, a third changes item subsets. This piecemeal approach cannot capture interactions between sources, nor can it predict how the relative importance of each source changes across evaluation settings. Leaderboard rankings built on single-condition evaluations are therefore sensitive to methodological choices that remain unreported and uncontrolled (Rodriguez et al., 2021; Dehghani et al., 2021). No study has simultaneously crossed infrastructure and methodology facets in a unified factorial design across benchmarks to decompose these sources, quantify their interactions, and compare the resulting variance structure.

Existing studies examine individual variance sources in isolation. Sclar et al. (2024) and BrittleBench (Romanou et al., 2026) documented 50–76% accuracy swings from prompt formatting; Yuan et al. (2025) showed up to 9% variation from numerical precision; Pezeshkpour and Hruschka (2024), Dodge et al. (2020), and Henderson et al. (2018) documented ordering and hyperparameter sensitivity—each holding other factors fixed. Messing (2026) applied G-theory to LLM evaluation across five methodology facets on a single benchmark (judge identity: 43.8%), but excluded infrastructure facets and cross-benchmark comparison. Madaan et al. (2024) proposed per-factor metrics without crossing factors. IRT approaches (Maia Polo et al., 2024; Lalor et al., 2019; Zhou et al., 2026) optimize item selection but ignore prompt, seed, and ordering facets. Bouthillier et al. (2021) established ANOVA-based variance decomposition for ML reproducibility. Three gaps remain: no study has (a) crossed infrastructure and methodology facets in a unified factorial design, (b) compared variance structures across response formats, or (c) translated variance components into design-dependent reproducibility budgets.

Through a factorial G-theory study (Shavelson

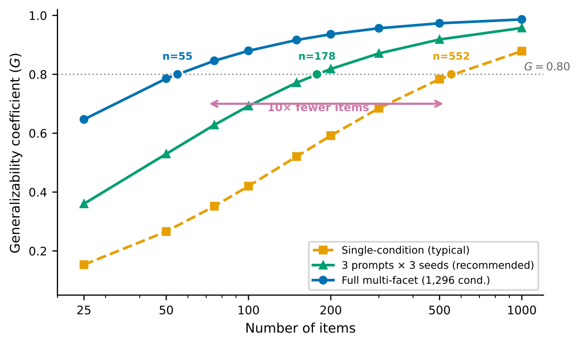

Figure 1: D-study projections for MMLU (single-model baseline). A recommended 3-prompt × 3-seed design reduces the budget from 552 to 178 items for $G \geq 0.80$; the full multi-facet design reduces further to 55 items.

and Webb, 1991; Brennan, 2001) crossing up to seven facets across eight open-weight 7–9B parameter models and five benchmarks (MMLU, ARC-Challenge, HellaSwag, GSM8K, MATH), totaling approximately 2.3 million records, we find that evaluation variance has a stable but format-dependent internal structure. D-study projections show that a recommended 3-prompt × 3-seed design achieves $G \geq 0.80$ with 178 items (1,602 evaluations per model), a 3.1× reduction from 552 single-condition items; the full 1,296-condition design further reduces to 55 items as a theoretical ceiling (Figure 1). Model×item interaction (up to 36%) indicates that models differ primarily in *which items* they find difficult (Pezeshkpour and Hruschka, 2024; Maia Polo et al., 2024).

Our contributions:

1. **Design-dependent reproducibility budgets.** A recommended 3-prompt × 3-seed design achieves $G \geq 0.80$ with 178 items (1,602 evaluations per model), a 3.1× reduction over single-condition designs (552 items); the full 1,296-condition design reduces to 55 items (71,280 evaluations). Under binary scoring, BF16 and FP16 require no separate control (<0.1% interaction). IRT item selection and G-theory protocol optimization compose: IRT-selected 116 items yield $G = 0.457$ single-condition but $G = 0.895$ multi-facet (Table 4).

2. **Format-associated variance patterns (exploratory).** MC benchmarks concentrate 68–72% of variance in item-related components; FF benchmarks disperse to 31–41% (item-level $d = 1.73$; benchmark-level confirmation limited by sample size). This hypothesis-generating observation warrants confirmation with larger benchmark samples.

We release an open-source G-theory pipeline and D-study budget calculator for evaluation design optimization.

2 Related Work

Variance Decomposition in ML/NLP Evaluation. Factorial decomposition of evaluation variance has roots in the ML reproducibility literature. Bouthillier et al. (2021) introduced an ANOVA-based template that decomposes benchmark variance across implementation details, hyperparameters, and data splits, establishing that design choices account for a substantial share of reproducibility failures. Subsequent studies applied similar decomposition to specific evaluation facets (Dodge et al., 2020; Henderson et al., 2018; Magnusson et al., 2023). Messing (2026) brought Generalizability theory into LLM evaluation, decomposing variance across five methodology facets on a single benchmark (judge identity: 43.8%; see Positioning below for detailed comparison). Madaan et al. (2024) proposed per-factor variance metrics for LLM evaluation benchmarks but did not cross factors simultaneously. Holistic evaluation frameworks (Liang et al., 2023; Kiela et al., 2021) standardize multi-scenario comparison but do not decompose variance sources. Blackwell et al. (2024) quantified uncertainty in benchmark scores, and Qian et al. (2026) conducted systematic meta-evaluation of LLM benchmarks; Burnell et al. (2023) called for rethinking evaluation reporting standards; Saxon et al. (2024) advocated model metrology as a framework for rigorous benchmark design. Recently, G-theory has been applied to LLM-based writing assessment (Song et al., 2025) and annotation reliability (Camuffo et al., 2026), but these studies address scoring consistency rather than benchmark evaluation variance. Kunievsky and Evans (2025) proposed an alternative decomposition that separates intent uncertainty, articulation variance, and model uncertainty, but this framework does not cross infrastructure and methodology facets in a factorial design. Our work extends these approaches by crossing seven facets across five benchmarks, enabling direct comparison of variance structures across evaluation formats.

Infrastructure and Prompt Variance. A parallel line of work examines individual variance sources in isolation. Yuan et al. (2025) documented up to 9% accuracy variation from numerical precision; Messina and Scotta (2026) and He and Thinking Machines Lab (2025) formalized and mitigated CUDA nondeterminism from floating-point non-associativity. On the methodology side, Mizrahi et al. (2024) advocated multi-prompt evaluation based on large-scale sensitivity analysis; Sclar et al. (2024) and BrittleBench (Romanou et al., 2026) found 50–76% accuracy swings from prompt formatting; Hua et al. (2025) showed that much of this sensitivity is an artifact of log-likelihood scoring (our generation-based protocol controls for this). Pezeshkpour and Hruschka (2024) and Alzahrani et al. (2024) documented ordering and benchmark sensitivity; Bui et al. (2025) and Biderman et al. (2024) documented seed and training sensitivity respectively. Each isolates one factor, precluding interaction estimation. Standardized evaluation frameworks such as lm-evaluation-harness (Gao et al., 2024) reduce implementation variance but do not control prompt, seed, or ordering facets.

IRT and Efficient Benchmarking. Item Response Theory (IRT) offers a complementary perspective on evaluation design by modeling item difficulty and discrimination parameters. Lalor et al. (2019) introduced IRT-based item analysis for NLP benchmarks, and Rodriguez et al. (2021) showed that IRT reveals hidden properties of evaluation items that raw accuracy scores obscure. Building on this foundation, Maia Polo et al. (2024) developed tinyBenchmarks, using IRT-based subsampling to reduce evaluation cost while preserving ranking accuracy, and Zhou et al. (2026) recommended a 116-item subset via PSN-IRT for efficient MMLU evaluation. Perlitz et al. (2023) and Kipnis et al. (2025) explored related strategies for benchmark compression; Li et al. (2025) extended IRT to adaptive testing. Complementary benchmark revision efforts address item quality directly: MMLU-Redux (Gema et al., 2025) identified annotation errors in MMLU, while MMLU-Pro (Wang et al., 2024) extended it with harder items and expanded option sets. These approaches optimize item selection or quality but do not model variance from prompt, seed, or ordering facets. IRT-selected items evaluated under a single methodological condition can therefore systematically underestimate the number of items required for reliable measurement across diverse evaluation settings.

Psychometrics Foundations. G-theory (Shavelson and Webb, 1991; Brennan, 2001; Cardinet et al., 2010) provides the canonical framework for multi-facet reliability analysis; Webb et al. (2006) detail the random versus fixed facet distinctions central to our model specification. Extensions to binary data include Mushquash and O'Connor (2006) and logistic-link generalized G-theory (Jiang and Skorupski, 2018; Tuerlinckx et al., 2006). We operationalize differential item functioning—established as a formal framework by Holland and Wainer (1993) and operationalized via statistical detection procedures by Clauser and Mazor (1998)—as model$\times$item interaction. Recent psychometric perspectives on LLM evaluation address construct validity (Kearns, 2026; Bean et al., 2025; Freiesleben, 2026) and measurement properties (Ye et al., 2025); Zhu et al. (2026) propose a benchmark health index for systematic meta-evaluation.

Positioning. Closest to our work, Messing (2026) applied G-theory to LLM evaluation through their TEE framework, decomposing five methodology facets on a single benchmark (judge identity: 43.8%). We extend this in three directions: (i) we fix the judge to the model itself, replacing judge identity with infrastructure facets and additional methodology facets (seed, ordering); (ii) we cross facets across five benchmarks, enabling cross-format comparison; and (iii) we translate variance components into reproducibility budgets via D-study projections. The two studies are complementary: TEE characterizes judge selection, while we characterize item-and-protocol selection. Relative to IRT approaches (Maia Polo et al., 2024; Zhou et al., 2026), G-theory optimizes an orthogonal dimension: conditions needed for target reliability rather than item discrimination.

3 Method and Experimental Design

3.1 Generalizability Theory

Generalizability theory (G-theory) treats each measurement as one observation from a universe of admissible conditions defined by the crossed facets of an evaluation design (Shavelson and Webb, 1991; Brennan, 2001). The *object of measurement* is the entity whose consistency is of interest. In our setting, test items (p) serve as the object of measurement, and the goal is to determine how reliably item-level scores generalize across the methodological conditions under which they are collected.

A *G-study* decomposes total observed variance into components σ^2_x for each facet x and their interactions. Each component quantifies how much variability that source contributes to the overall measurement. Large interaction components (e.g., σ^2_{pM} for item-by-model) indicate that facets do not affect items uniformly but instead create differential difficulty profiles. We estimate variance components via Henderson Method I, an unbiased ANOVA-based estimator for balanced designs (Brennan, 2001, Ch. 3), with 1,000-resample bootstrap confidence intervals (Efron and Tibshirani, 1993) (item-level resampling with replacement, preserving the fully-crossed structure across all other facets) (Li, 2023).

A *D-study* projects the generalizability coefficient under hypothetical designs with different numbers of conditions per facet:

$$G_{\text{item}} = \frac{\sigma^2_p}{\sigma^2_p + \sigma^2_{\text{rel}}} \tag{1}$$

where $\sigma^2_{\text{rel}} = \sum_i \sigma^2_{p\times f_i}/n_{f_i} + \sigma^2_{\text{residual}}/\prod_i n_{f_i}$ aggregates all item-by-facet interaction variances, each divided by the number of conditions for the corresponding facets (Brennan, 2001, Ch. 4). A D-study projects reliability for a test averaging over n_p items via

$$G(n_p) = \frac{n_p\,\sigma^2_p}{n_p\,\sigma^2_p + \sigma^2_{\text{rel}}} \tag{2}$$

enabling independent verification of all budget projections in Table 4. This prospective capability distinguishes G-theory from classical ANOVA: rather than only attributing variance post hoc, a D-study enables optimization of evaluation designs *before* data collection (Shavelson et al., 1989). We define *item-related variance* as the sum of the item main effect (σ^2_p) and the model×item interaction (σ^2_{pM}). The latter captures differential item functioning (DIF): items whose difficulty ranking changes across models (Brennan, 2001, Ch. 4). Because DIF is operationally an item-level property—it reflects model-specific item difficulty profiles rather than a global model ability shift—aggregating $\sigma^2_p + \sigma^2_{pM}$ provides a unified measure of item-driven variance. We note a semantic difference from educational measurement: person×item DIF reflects unidirectional person responses to items, whereas model×item DIF in LLM evaluation simultaneously reflects architecture-specific capability profiles—a dual interpretation. Bundling $\sigma^2_p + \sigma^2_{pM}$ has psychometric justification (Brennan, 2001, Ch. 4), and we report unbundled components throughout to ensure transparency.

In ML terms, a G-study is a multi-way ANOVA where each facet is a design variable and variance components quantify each variable's contribution; a D-study projects the number of conditions per facet needed for a target reliability G, analogous to a signal-to-noise ratio where signal is item difficulty variance and noise is everything that changes across evaluation conditions.

Throughout this paper, we use the following notation: p = item, M = model, π = numerical precision, δ = prompt template, s = random seed, τ = sampling temperature, o = answer ordering, and σ^2_x = variance component for source x.

3.2 Experimental Design: MMLU

We design a fully crossed factorial experiment (exp-001) that simultaneously varies seven facets of LLM evaluation on MMLU. Table 1 summarizes the complete design matrix, and Figure 2 illustrates the crossing structure.

Facets. Items (p): 200 MMLU questions selected via purposive stratified sampling across 10 of MMLU's 57 subjects (20 per subject, deterministic selection with arbitrarily chosen seed 42). The 10 subjects span four disciplinary domains—STEM: abstract algebra, college physics, high school mathematics, computer security; humanities: world religions, philosophy; social sciences: US foreign policy, international law; professional: professional medicine, clinical knowledge—ensuring that variance estimates are not dominated by a single knowledge domain. Resampling stability (CV = 3.1%) was evaluated within these 10 subjects; cross-subject generalization remains untested. Models (M): eight open-weight 7–9B parameter instruction-tuned models spanning eight architecture families—DeepSeek-7B, Gemma-2-9B, InternLM2.5-7B, Llama-3.1-8B-Instruct, Mistral-7B-v0.3-Instruct, OLMo-2-7B, Qwen3-8B, and Yi-1.5-9B (full identifiers in Appendix H)—treated as a quasi-random facet (§3.6). Numerical precision (π): BF16, FP16, and FP32, all evaluated with `enforce_eager=True` for fair comparison across precision levels. Prompt template (δ): six surface perturbations of the evaluation instruction (Sclar et al., 2024), varying option layout (lettered vs. numbered), instruction wording (direct vs. contextual), and answer prompt cue for MC bench-

Table 1: Fully crossed G-theory design. Exp-001 evaluates MMLU across all facet levels; exp-002 extends to five benchmarks with reduced crossing. †BF16 balanced subset (2 temperatures: 0.0, 0.7) for cross-model analysis; the full single-model design uses all 3 levels.

Facet	Levels	exp-001	exp-002
Item (p)	Stratified sample	200	200×5
Model (M)	8 architecture families	8	8
Precision (π)	BF16 / FP16 / FP32	3	1
Prompt (δ)	Paraphrases	6	6
Seed (s)	Fixed set	6	6
Temperature (τ)	0.0 / 0.3 / 0.7	3	2
Ordering (o)	Demo permutations	4	4
Total records		460,800†	2,304,000

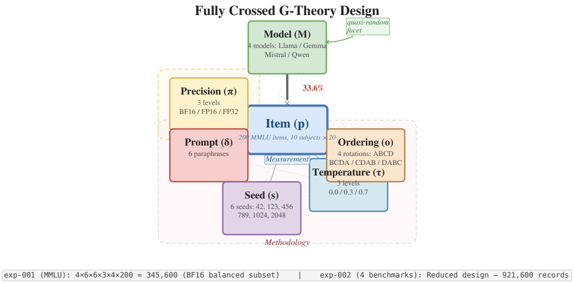

Figure 2: Fully crossed G-theory design for LLM evaluation. Each observation is a binary correctness score determined by the crossing of seven facets: item (p), model (M), precision (π), prompt (δ), seed (s), temperature (τ), and ordering (o).

marks; for FF benchmarks, templates vary chain-of-thought elicitation strategy (full template text in Appendix H). Random seed (s): six fixed values (42, 123, 456, 789, 1024, 2048). Sampling temperature (τ): three levels (0.0, 0.3, 0.7) spanning deterministic to moderately stochastic decoding. Answer ordering (o): four permutations of few-shot demonstration examples (for MMLU; structurally inert for other benchmarks, see §3.3) (Pezeshkpour and Hruschka, 2024).

Sample sizes. The cross-model BF16 balanced subset yields 460,800 records (8 models × 200 items × 6 prompts × 6 seeds × 2 temperatures × 4 orderings). A single-model analysis (Llama) spans all three precisions, yielding 259,200 binary observations across 1,296 fully balanced conditions. The BF16 subset serves as the primary cross-model sample because BF16 is the standard inference precision for these architectures.

3.3 Cross-Benchmark Extension

To test whether the variance structure observed on MMLU generalizes across evaluation formats, exp-002 extends the design to five benchmarks: MMLU, ARC-Challenge, and HellaSwag (multiple-choice), and GSM8K and MATH (free-form generation). The design crosses 8 models × BF16 × 2 temperatures (0.0, 0.7) × 6 prompts × 6 seeds × 4 orderings × 200 items per benchmark, totaling 2,304,000 records. This reduced crossing fixes precision at BF16 and uses two temperature levels to concentrate statistical power on format-associated differences. The ordering facet operates only on MMLU, where it selects among four permutations of the few-shot demonstration examples. For all other benchmarks, the prompt construction does not use the ordering parameter; ordering is structurally inert ($\sigma^2_{\text{ordering}} = 0$ by construction) and is retained solely for design balance. The two temperature levels bracket the most common evaluation configurations: deterministic ($\tau = 0.0$) and a moderate stochastic setting ($\tau = 0.7$) representative of practical deployment. A subset analysis confirms that this reduced crossing preserves the variance structure of the full design ($\Delta G = 0.0008$; Appendix A).

3.4 Evaluation Protocol

All evaluations use generation-based scoring (free-text response with answer extraction) rather than log-likelihood ranking, following Hua et al. (2025). Binary correctness (1/0) is the primary metric; text-exact-match is secondary. Inference uses vLLM 0.9.2 without batch-invariant kernels, preserving CUDA-level nondeterminism as a measurable variance source (Messina and Scotta, 2026; He and Thinking Machines Lab, 2025). Few-shot protocols: 5-shot MMLU/ARC, 4-shot MATH, 8-shot GSM8K, 10-shot HellaSwag.

3.5 Binary Data Assumptions

Standard G-theory assumes continuous, approximately normal outcome data (Shavelson and Webb, 1991). Our binary correctness scores (0/1) violate this assumption. We establish robustness through simulation and estimator properties, with two additional analyses (GLMM complement, accuracy-range stability) in Appendix A.

Henderson Method I produces unbiased estimates for balanced designs regardless of outcome distribution (Brennan, 2001, Ch. 12), and REML estimates differ by <0.5pp on all components. A probit simulation, GLMM complement ($\rho = 1.0$), and accuracy-range analysis all confirm robustness (Appendix A).

All findings in this paper are conditioned on binary correctness scoring. This measurement resolution collapses continuous generation quality into a pass/fail outcome, which may amplify item-related variance by reducing within-item response variability and mask finer-grained variance sources (e.g., partial credit, reasoning quality) that would emerge under richer scoring rubrics. The text-exact-match comparison in §4.1 ($G = 0.007$ vs. 0.936) illustrates how the scoring metric can fundamentally reshape variance structure. A partial-credit analysis on GSM8K (8 models, step-level scoring) confirms this conditionality: G_{item} drops from 0.86 to 0.77, with all eight models showing $G_{\text{partial}} < G_{\text{binary}}$ (mean $\Delta G = -0.135$; Appendix A), providing evidence that D-study budgets have limited transferability to finer-grained scoring rubrics.

3.6 Model Facet Specification

Our target universe comprises open-weight, instruction-tuned models in the 7–9B parameter range (eight architecture families); results may not generalize to API-based, base, or >9B models. Our eight purposively selected models are treated as a quasi-random facet (Meredith, 1993; Putnick and Bornstein, 2016). We report G under random-model assumptions, the more conservative specification: across five benchmarks, treating models as fixed yields G_{fixed} of 0.951–0.990 (mean 0.975) versus G_{random} of 0.863–0.896 (mean 0.882), a consistent increase of 0.077–0.113 (mean 0.093), because model×item interaction (30–36%) enters the relative error variance under random assumptions. Consequently, our D-study budgets are upper bounds—practitioners will not under-sample regardless of facet classification. The variance decomposition itself (Tables 2–3) is invariant to this choice. Leave-one-model-out analysis preserves component rankings across all five benchmarks, and Cohen's κ pairwise agreement (mean $\kappa = 0.33$–0.46) confirms genuine differential item functioning (Appendix A). Four-model subset consistency checks match the observed eight-model G within mean $|\Delta G| = 0.035$ across all five benchmarks (§4.5).

For reference, G_{model} (treating models as the object of measurement) under the recommended 3-prompt × 3-seed design with 200 items averages 0.79 across five benchmarks (range: 0.52 on ARC to 0.93 on HellaSwag). Model ranking reliability depends primarily on model×prompt interaction—ARC (8.0%) and GSM8K (8.5%) show lower G_{model} because prompt sensitivity reduces model discriminability. G_{item} remains the primary metric because item reliability is the direct optimization target for evaluation design; model reliability is a downstream consequence that improves with item count.

4 Experiments

All experiments use open-weight 7–9B parameter models evaluated with vLLM 0.9.2 on a single NVIDIA RTX PRO 6000 GPU (96GB) with binary correctness scoring.

4.1 Single-Model Baseline

We first establish the variance structure within a single model (Llama-3.1-8B-Instruct) on MMLU to isolate methodological facet contributions before introducing cross-model variation.

Design. The G-study crosses six facets: 200 items × 3 numerical precisions (BF16, FP16, FP32) × 6 prompt templates × 6 random seeds × 3 temperature levels (0.0, 0.3, 0.7) × 4 answer orderings, yielding 259,200 binary observations. Variance components are estimated via Henderson Method I with bootstrap confidence intervals (1,000 resamples).

Results. Under binary scoring, item difficulty (`item_id`) accounts for 58.4% of total variance, with the residual term contributing 23.1%. All main facet effects are negligible (<1% each): precision 0.9%, prompt template, temperature, seed, and ordering each <0.5%. The interesting structure appears in item-conditional interactions: prompt×item is the largest at 5.5%, followed by seed×item (4.7%), precision×item (3.3%), temperature×item (2.9%), and ordering×item (0.8%). This pattern indicates that methodological choices do not shift performance uniformly across items; instead, specific items are differentially sensitive to specific facets—prompt template variation produces the largest item-conditional effect.

The resulting generalizability coefficient is $G_{\text{item}} = 0.936$, indicating that 200 items under multi-facet conditions produce highly reliable rankings. A D-study projection shows $G \geq 0.80$ is achievable with as few as 55 items in this multi-facet design.

For text-exact-match scoring (requiring verbatim string matches), the variance structure shifts dramatically: temperature dominates at 79.7% of variance, and G_{item} collapses to 0.007. This metric-

Table 2: Variance decomposition for the cross-model MMLU design (8 models, BF16, 460,800 records). Components estimated via Henderson Method I. $G_{\text{item}} = 0.896$.

Component	Var.%
item_id	**37.9**
model×item	30.2
residual	25.0
model	2.8
model×prompt	1.6
prompt×item	0.9
temp.×item	0.6
seed×item	0.5
ordering×item	0.3
all others (each)	<0.1

conditional divergence motivates our exclusive use of binary correctness for the remaining analyses.

4.2 Cross-Model Variance

Extending to eight models (BF16 subset, 460,800 records), Table 2 shows that item difficulty drops from 58.4% to 37.9%, while model×item interaction emerges at 30.2% (up to 36% across benchmarks, §4.3). The model main effect is 3–5% (MC) and 9–11% (FF)[1], confirming that models differ primarily in *which items* they find difficult rather than in overall ability level (contamination implications in §6).

4.3 Exploratory: Format-Associated Variance Patterns

We extend to five benchmarks (three MC: MMLU, ARC-Challenge, HellaSwag; two FF: GSM8K, MATH) using exp-002 (8 models × BF16 × 2 temperatures × 6 prompts × 6 seeds × 4 orderings × 200 items). Table 3 reveals a format-associated pattern: MC benchmarks concentrate item-related variance (item + model×item DIF) at 68–72%, while FF benchmarks disperse it to 31–41% ($n = 5$; permutation $p = 0.10$; item-level $d = 1.73$, benchmark-level CI spans zero); we report this as a descriptive observation. This does not extend to item budgets: HellaSwag (MC, 118 items for $G \geq 0.80$) exceeds ARC (98) and MATH (101), showing format alone does not predict evaluation cost (Figure 3). The factorial design is not fully symmetric: ordering applies only to MMLU, and prompt templates serve format-specific functions, potentially confounding format comparisons. Per-benchmark details—including HellaSwag and ARC exceptions, FF residual analysis (Table 11), difficulty dispersion as a variance predictor (Figure 5; Table 10)—are in Appendix D. All format-associated findings are conditional on binary scoring (§3.5).

[1]Ranges rounded to nearest integer; exact values are 2.8–5.4% (MC) and 8.5–10.7% (FF) per Table 3.

Table 3: Format-associated variance structure across five benchmarks (exp-002, 8 models). Variance components (%) per benchmark; MC = multiple-choice, FF = free-form. Bold = dominant non-residual component. †For MC, reports prompt×item interaction (prompt main effect <1%); for FF, reports prompt main effect (item-conditional interactions absorbed into residual).

	MC			FF	
	MMLU	ARC	Hella.	GSM	MATH
item	**37.9**	**37.9**	34.3	16.9	24.0
mod.×item	30.2	33.7	**36.0**	14.6	17.3
model	2.8	3.1	5.4	10.7	8.5
prompt†	0.9	1.4	2.1	5.4	0.8
mod.×prmpt	1.6	8.0	0.2	8.5	1.7
residual	25.0	15.4	21.4	39.9	43.9
all others	1.6	0.5	0.6	4.0	3.8

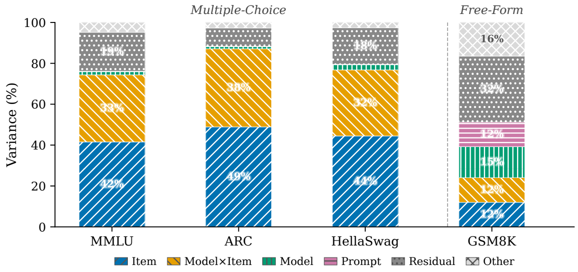

Figure 3: Variance decomposition (Henderson Method I) across five benchmarks and eight models. Multiple-choice benchmarks share an item-dominated structure; HellaSwag is the exception where model×item (36.0%) exceeds item difficulty (34.3%). Free-form benchmarks (GSM8K, MATH) distribute variance across item, model, and interaction components with no single dominant source.

4.4 Precision Separability

Under binary scoring, BF16 and FP16 are functionally interchangeable (<0.1% interaction (Yuan et al., 2025)); under text-exact-match, a precision×temperature interaction emerges at 1.3%, consistent with §4.1. Precision need not be controlled for binary evaluation at BF16/FP16 (Shanmugavelu et al., 2024).

4.5 Design-Dependent Reproducibility Budgets

How many items does a given evaluation design need to achieve a target reliability? We translate variance components into item budgets via D-study

Table 4: D-study reproducibility budgets under different evaluation designs (single-model, MMLU). Total evals = items × conditions.

Design	Cond.	$n_{G\geq.80}$	Total	G_{200}	G_{116}
Multi-facet	1,296	**55**	71,280	**.936**	—
Single-cond.	1	552	552	.592	.457
IRT+multi	1,296	—	150,336	—	.895‡
IRT baseline	—	116	116	—	.457†

† PSN-IRT 116 items (Zhou et al., 2026) under single-condition: $G = 0.457$.
‡ Same 116 items under multi-facet: $G = 0.895$.

projections.

Table 4 compares evaluation designs at the $G \geq 0.80$ threshold using single-model (Llama-3.1-8B-Instruct) MMLU variance components. The multi-facet design (1,296 conditions per item) achieves $G = 0.936$ at 200 items and requires only 55 items for $G \geq 0.80$. A single-condition design reaches only $G = 0.592$ at 200 items and requires 552 items—a 10.0× budget ratio stable across subset sizes (CV = 1.9%).

Zhou et al. (2026)'s IRT-selected 116 items yield $G = 0.457$ under single-condition but $G = 0.895$ under multi-facet conditions, demonstrating that IRT item selection and G-theory protocol optimization are orthogonal and complementary. An internal consistency check using four-model subsets confirms projection stability ($|\Delta G| \leq 0.029$ for MC, ≤ 0.065 for FF; Table 8); the corresponding ∼23% budget increase is consistent across all five benchmarks (Figure 4).

For practitioners, the design choice depends on the evaluation bottleneck: the full 1,296-condition design minimizes items (55 items, 71,280 evaluations per model); the recommended 3-prompt × 3-seed design balances cost and reliability (178 items, 1,602 evaluations)[2]; single-condition designs require 552 items at minimal per-item cost. Multi-facet designs suit settings where item construction is expensive (e.g., expert-crafted benchmarks); more items under fewer conditions suit compute-constrained evaluation (adoption gradient in Appendix C). Cross-model projections show budget ratios of 11–22× across benchmarks (bootstrap CIs all exceed 10×; Appendix B), with multi-facet budgets of 93–128 items (95% bootstrap CIs spanning 71–162; Table 7).

[2]The additional conditions require approximately 8 GPU-minutes per model across five benchmarks on a single accelerator, making the multi-facet overhead negligible relative to its diagnostic value.

Projection assumptions. Cross-model budgets increase to 93 items because model×item interaction enters relative error variance. Bootstrap CIs (71–162 items) and LOMO analysis (Appendix A) assess stability; G_{item} is stable across benchmarks (0.863–0.896, CV = 1.4%; Appendix B). All projections assume the 2024–2025 model generation; the 72B extension (§4.6) provides preliminary support.

4.6 Preliminary Scale Validation

Within-family analysis (Qwen2.5 3B/7B/72B on MMLU) shows qualitatively invariant variance structure: item difficulty dominates at all three scales (73–83%, all $G > 0.95$) with consistent facet ranking. The proportional increase at 72B is consistent with the Bernoulli variance ceiling: total variance (0.113) matches $p(1-p)$ at grand mean 0.87, compressing all components while item difficulty—the hardest to eliminate—retains the largest share. A crossed 3B×7B×72B design yields model×item = 38.5% > item = 35.7%, reflecting expected DIF amplification across a 68–87% accuracy range ($G_{\text{item}} = 0.720$; three models insufficient). The out-of-sample 72B provides preliminary within-family evidence that variance structure is stable across a 24× parameter range within Qwen2.5. A cross-family check with Llama-3.1-70B-Instruct (same protocol, BF16, TP=4) yields item difficulty = 74.5% ($G_{\text{item}} = 0.957$), confirming the item-dominated structure across architecturally distinct families: all main effects are zero, and the facet ranking matches Qwen2.5-72B. The 8.7pp item share gap (74.5% vs. 83.2%) reflects accuracy-dependent Bernoulli ceiling compression, not structural divergence.

Robustness checks. Three MMLU resamples yield CV = 3.1%; treating models as fixed increases G by 0.077–0.113, confirming conservative budgets (§3.6; seven additional analyses in Appendix A).

5 Conclusion

A recommended 3-prompt × 3-seed design reduces item requirements from 552 to 178 for $G \geq 0.80$, confirmed across five benchmarks under binary scoring; multi-facet designs minimize items when item construction is costly, while single-condition designs suit compute-constrained settings. The variance structure is not format-invariant: MC benchmarks concentrate 68–72% of variance in item-related components, while FF benchmarks disperse to 31–41%, suggesting that reproducibility budgets should be calibrated per benchmark format. G_{item} and G_{model} have different bottlenecks—item count for the former, prompt standardization for the latter—providing orthogonal optimization paths: adding items improves item reliability, while controlling prompt and seed conditions improves model discriminability. All budgets assume binary scoring; extending to other model scales and continuous metrics is a priority. We recommend that benchmark designers adopt multi-facet validation (at minimum 3 prompts $\times$ 3 seeds) as standard practice when sizing pilot studies, and that leaderboard operators average over prompt and seed conditions to improve ranking stability. Code, data, and a D-study calculator will be released upon publication.

6 Limitations

Our eight models share the 2024–2025 instruction-tuning paradigm; temporal correlation may limit exchangeability, and results do not generalize to API-based or base models. Preliminary scale validation (§4.6) with Qwen2.5 (3B/7B/72B) and cross-family Llama-3.1-70B confirms consistent variance structure across two architecture families, but validation beyond these two families is needed before extrapolating budget recommendations to arbitrary architectures.

The benchmark sample ($n = 5$) limits statistical power for format-associated observations ($p = 0.10$); other formats (code generation, multi-turn dialogue) may differ. All D-study budgets reported in this paper are strictly valid only under binary correctness scoring. Our partial-credit analysis (GSM8K, 8 models) shows G drops from 0.86 to 0.77 (crossed design) with all models showing $G_{\text{partial}} < G_{\text{binary}}$, suggesting limited transferability to finer-grained scoring rubrics; the analysis is limited to a single benchmark, and its generalizability to other tasks is unknown. Our temperature range (0.0–0.7) does not cover high-temperature regimes ($\tau > 1.0$) sometimes used in generation tasks. Our design varies few-shot example ordering but not example identity, a known variance source. Results depend on vLLM 0.9.2; different inference engines or vLLM versions may alter CUDA-level nondeterminism patterns.

All eight models may have encountered benchmark items during pre-training; differential contamination across architectures could inflate model$\times$item interaction (Golchin and Surdeanu, 2023; Oren et al., 2024), though consistent variance structures across three independently constructed MC benchmarks provide indirect evidence against contamination as the sole driver. Contamination affects absolute budget estimates but has limited impact on the multi-facet vs. single-condition budget *ratio*, since both designs are equally exposed to contaminated items.

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A Additional Robustness Analyses

We report nine robustness analyses that verify the stability of the variance decomposition under alternative analytical choices.

GLMM complement. A generalized linear mixed model (GLMM) with marginal logit link produces variance component rankings that perfectly preserve all seven grouped component orderings ($\rho = 1.0$) relative to Henderson Method I (Jiang and Skorupski, 2018; Tuerlinckx et al., 2006). This confirms that the linear variance decomposition is robust to the binary outcome distribution.

Accuracy-range stability. The `item_id` variance share remains stable across subsets with mean accuracy from 0.62 to 0.72 ($r = -0.048$). This rules out distortion from binary floor or ceiling effects at observed accuracy levels.

Estimator robustness. We compare Henderson Method I with restricted maximum likelihood (REML-EM, 500 iterations, practical convergence $\Delta < 0.003$ percentage points). All variance components differ by <0.5 percentage points between estimators. The largest discrepancy occurs on the residual component (0.47pp); all substantive components (item, model×item, model) differ by <0.3pp. Henderson Method I is preferred for our balanced designs because it produces unbiased estimates without distributional assumptions (Brennan, 2001, Ch. 12).

Probit simulation. A probit simulation matched to our 1,296-condition design confirms that Henderson Method I recovers variance component rankings from dichotomized data (Spearman $\rho = 0.732$ between binary and continuous rankings, excluding residual; `item_id` dominant in all 100 replications; top-5 ordering correct in 79% of replications). Bias analysis by difficulty stratum shows $|\text{bias}| = 0.28$–0.37 for items with $p \in [0.5, 0.8]$, where the Bernoulli variance ceiling $p(1-p)$ is largest. This analysis covers the observed accuracy range ($p \in [0.3, 0.9]$); extreme-difficulty items ($p < 0.3$ or $p > 0.9$) are rare in our sample (<8% of items) and may exhibit larger deviations. Henderson Method I absorbs this Bernoulli-inherent variance into the residual, explaining inflated bias near medium difficulty; systematic component ordering is robustly recovered across all replications. Binary scoring inflates residual variance by absorbing the Bernoulli $p(1-p)$ ceiling, which enters σ^2_{rel} and increases D-study item requirements. Within a matched probit simulation, binarization increases D-study budgets by approximately 36% ($99 \rightarrow 135$ items for $G \geq 0.80$), driven by residual inflation ($16\% \rightarrow 41\%$) and item variance compression ($50\% \rightarrow 32\%$). Facet×item interactions—the components entering σ^2_{rel}—are best preserved under binarization (MAPE = 12.8%, range 8.7–17.5%), while residual variance inflates as expected from Bernoulli $p(1-p)$ noise (MAPE = 156%). D-study budgets derived from binary data are therefore conservative: continuous-scored designs would require equal or fewer items for the same G threshold. Null calibration (ratio $= 1.0008$) confirms that this conservatism does not distort relative component magnitudes.

CI method comparison. Bootstrap (1,000 item-level resamples with replacement) and delete-1 jackknife confidence intervals agree closely: width ratios range from 0.96 to 1.04 across all variance components. Bootstrap CIs capture item-sampling variability conditional on these 8 specific models; model-level stability is assessed separately via LOMO (below). The agreement indicates that the dependence structure introduced by item-level bootstrap resampling (which creates ∼127 unique items per 200-item resample) does not materially distort interval estimates. Stratified resampling by subject (10 subjects × 20 items) produces CIs within 4% of simple item-level bootstrap, confirming that the within-subject dependence structure does not materially affect interval estimates.

Facet specification sensitivity. Treating temperature and ordering as fixed (rather than random) facets changes G_{item} by only 0.37 percentage points (all-random: 0.820 vs. fixed temperature+ordering: 0.824). The small difference reflects the negligible temperature×item (0.65%) and ordering×item (0.54%) variance shares. The random-facet assumption is therefore robust to facet classification.

Leave-one-model-out (LOMO). Across all 40 leave-one-model-out subsets (8 models × 5 benchmarks), G_{item} ranges from 0.833 to 0.919, with CV $\leq$ 1.25% per benchmark. No single model removal changes G by more than 0.030 (GSM8K, removing Gemma-2-9B). Item-related variance rankings are preserved in all 40 subsets. Table 5 reports per-benchmark stability. This supports treating models as exchangeable within the target universe of 7–9B instruction-tuned models (§3.6).

Table 5: LOMO stability across five benchmarks (8 models × 5 benchmarks = 40 subsets).

Benchmark	G_{full}	CV (%)	Max $\lvert\Delta G\rvert$	Most impactful
MMLU	0.896	0.49	0.019	InternLM2.5-7B
ARC	0.892	0.94	0.026	Llama-3.1-8B
HellaSwag	0.872	1.09	0.028	Llama-3.1-8B
GSM8K	0.863	1.25	0.030	Gemma-2-9B
MATH	0.888	1.20	0.029	InternLM2.5-7B

Inter-model agreement. We computed pairwise Cohen's κ and Fleiss' κ across all model pairs on each benchmark. Mean pairwise κ ranges from 0.326 (MATH) to 0.461 (MMLU); Fleiss' κ ranges from 0.309 (GSM8K) to 0.456 (MMLU). HellaSwag uses six models ($\kappa = 0.332$) due to missing OLMo and Yi data. The fair-to-moderate agreement independently confirms that the model×item interaction (30–36% of variance) reflects genuine differential item functioning: models disagree substantially on which items are difficult, consistent with prior DIF findings in educational measurement (Holland and Wainer, 1993; Clauser and Mazor, 1998).

Scoring metric sensitivity. To probe metric conditionality, we compared binary correctness with step-level partial credit on GSM8K using the full 8-model crossed design (200 items, 2,304 conditions per item). Under partial credit, G_{item} drops from 0.863 to 0.768, with residual variance rising from 39.9% to 64.9% and item variance falling from 16.9% to 10.3%. All eight models individually show $G_{\text{partial}} < G_{\text{binary}}$ (mean $\Delta G = -0.135$, range -0.025 to -0.309), with ΔG magnitude moderated by model accuracy: higher-accuracy models have fewer incorrect responses whose scores shift under partial credit (e.g., OLMo at 70.1% accuracy shows $\Delta G = -0.025$, while Llama at 49.0% shows $\Delta G = -0.309$). The mechanism is clear: 48.1% of incorrect responses have numerically correct arithmetic steps but incorrect problem setup, receiving partial score $=$ 1.0 despite binary score $= 0$—this compresses item discrimination and inflates residual variance. Binary scoring amplifies prompt sensitivity by collapsing reasoning quality into pass/fail, confirming that all findings in this paper are metric-conditional.

Facet specification sensitivity: random vs. fixed model. Table 6 reports per-benchmark generalizability coefficients under random and fixed model specifications. The consistent ΔG of 0.077–0.113 confirms that random-model budgets are strictly more conservative across all benchmarks.

Table 6: Generalizability coefficients under random vs. fixed model facet specification across five benchmarks.

Benchmark	G_{random}	G_{fixed}	ΔG
MMLU	0.896	0.984	+0.088
ARC	0.892	0.990	+0.098
HellaSwag	0.872	0.984	+0.113
GSM8K	0.863	0.951	+0.089
MATH	0.888	0.965	+0.077
Mean	0.882	0.975	+0.093

Null calibration. We generate 100 Bernoulli-distributed datasets with item difficulties matched to the empirical distribution but no systematic facet effects. The observed-to-null variance ratio is 1.0008, confirming that Henderson Method I does not introduce systematic bias when applied to binary outcomes. The null simulation also verifies that the item variance share (58.4%) is not an artifact of the Bernoulli $p(1-p)$ ceiling: the null produces item variance of 0.08%, three orders of magnitude below the observed value. Across all 100 replications, the same three components (item, model×item, residual) emerge as the top three (set consistency 100%), with Kendall's $\tau = 0.81$ between binary and continuous rankings.

Reduced crossing validation. To verify that the reduced crossing in exp-002 (BF16 only × 2 temperatures) does not introduce confounding, we extracted the corresponding subset from the fully crossed exp-001 design. The generalizability coefficient differs by only 0.0008 ($G = 0.9361$ vs. 0.9369), and all non-precision variance components differ by less than 2 percentage points. The 10pp increase in item variance share (58.4% → 68.5%) is fully explained by the removal of precision-related components (∼4.5% total), which redistributes across remaining sources. This confirms that the exp-002 design preserves the variance structure of the full crossing.

B Cross-Model D-Study Projections

Table 7 extends the single-model D-study (Table 4 in main text) to per-benchmark cross-model projections with bootstrap confidence intervals (1,000 item-level resamples, 8 models, 2,304 conditions per item). The efficiency ratio is format-associated: MC benchmarks show 11–13× (mean 12.0× [11.6, 12.5]), while FF benchmarks show 21–22× (mean 21.5× [20.4, 22.8]).

Table 7: Cross-model D-study reproducibility budgets with 95% bootstrap CIs. $n_{G\geq 0.80}$: items needed. Multi = multi-facet (2,304 cond.); Total = items × 2,304. Single = single-condition (total evals = item count).

	Multi-facet (2,304 cond.)			Single-cond.		
	Items	95% CI	Total	Items	95% CI	Ratio [95% CI]
Multiple-choice						
MMLU	93	[76, 116]	214K	1,216	[1004, 1491]	13.1 [12.5, 13.7]
ARC	98	[71, 139]	226K	1,078	[793, 1512]	11.0 [10.7, 11.4]
Hella.	118	[93, 151]	272K	1,403	[1120, 1790]	11.9 [11.5, 12.3]
Free-form						
GSM8K	128	[103, 162]	295K	2,768	[2230, 3531]	21.6 [20.7, 23.0]
MATH	101	[86, 122]	233K	2,160	[1853, 2631]	21.4 [20.2, 22.5]

The higher item requirements relative to the single-model analysis (55 items in Table 4) reflect the additional model×item interaction variance that enters σ^2_{rel} when models are crossed as a facet. The format-associated ratio pattern—FF benchmarks showing ∼2× the MC ratio—is consistent with the distributed variance structure observed in §4.3, where FF benchmarks allocate more variance to model and model×prompt components.

Internal consistency check. Table 8 reports consistency checks from four exp-001 models (DeepSeek, Llama, Mistral, Qwen3) to the full eight-model design across all five benchmarks. MC benchmarks show $|\Delta G| \leq 0.029$ (<3.5% error), while FF benchmarks show larger deviations ($|\Delta G| \leq 0.065$, ∼7%) due to limited model diversity in the four-model subset. FF predictions consistently underestimate actual G (conservative); MC predictions show mixed directions with small deviations ($|\Delta G| \leq 0.029$).

Table 8: Internal consistency: four-model subset vs. eight-model G_{item}. FF subsets underestimate actual G (conservative); MC shows mixed directions with small deviations.

Benchmark	Predicted G	Actual G	$\lvert\Delta G\rvert$	Error%
Multiple-choice				
MMLU	0.888	0.896	0.009	0.96
ARC	0.904	0.892	0.012	1.38
HellaSwag	0.901	0.872	0.029	3.33
Free-form				
GSM8K	0.803	0.863	0.059	6.88
MATH	0.823	0.888	0.065	7.30
Mean	—	—	0.035	3.97

Budget sensitivity to estimation error. Figure 4 shows the effect of $|\Delta G| = 0.035$ estimation error on item budgets across all five benchmarks. The percentage increase is remarkably stable (23.0–24.5%, mean 23.2%), because the relative budget adjustment $\Delta n/n$ depends on the G-formula ratio rather than on absolute variance components. Absolute Δn ranges from 13 (ARC) to 67 (GSM8K), scaling with the baseline budget.

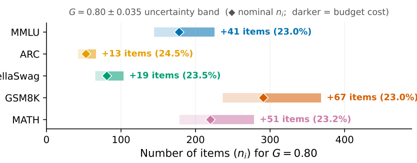

Figure 4: Effect of $|\Delta G| = 0.035$ estimation error on item budgets across five benchmarks. The ∼23% budget increase is consistent across all benchmarks, indicating D-study budgets are robust to estimation uncertainty.

Cross-benchmark G stability. G_{item} values cluster in a narrow band across all five benchmarks (0.863–0.896, CV = 1.4%), despite qualitatively different variance structures: MC benchmarks concentrate 68–72% in item-related components while GSM8K disperses variance across item (17%), model (11%), and interaction (15%) sources (Table 3). This stability implies that D-study budget recommendations calibrated on one benchmark provide reasonable order-of-magnitude guidance for others. However, the 0.033 range in G_{item} translates to a 71–128 item range for multi-facet $G \geq 0.80$ (Table 7), and the variance compositions differ substantially—benchmark-specific G-studies remain necessary to identify dominant variance sources and optimize condition selection.

C Intermediate D-Study Designs

Table 9 shows how reliability improves as evaluation designs increase from single-condition to full multi-facet crossing. The steepest gain occurs at the "minimal upgrade" level (2 prompts): G_{200} jumps from 0.592 to 0.692 at 2× cost. A minimal sufficient design for $G \geq 0.80$ is 3 prompts × 3 seeds (9 conditions, $G_{200} = 0.818$, $n_{G\geq0.80} = 178$). Beyond the "standard" level (32 conditions), adding orderings provides negligible benefit (ordering×item <1%), confirming that prompt and seed diversity are the primary reliability drivers.

Table 9: Intermediate D-study designs (single-model, MMLU). Designs are ordered by increasing conditions per item; G_{200} and $n_{G\geq0.80}$ are D-study projections from exp-001 variance components.

Design	Conditions	G_{200}	$n_{G\geq.80}$
Single-condition	1	.592	552
Minimal upgrade (2p)	2	.692	357
Easy upgrade (2p×2s)	4	.766	245
Moderate (3p×3s)	9	.818	178
Standard (4p×4s×2t)	32	.869	121
Enhanced (+4 orderings)	72	.871	119
Full multi-facet	1,296	.936	55

D Format-Associated Variance: Per-Benchmark Details

This section supplements the compressed format-associated analysis in §4.3 with per-benchmark details.

MC benchmark exceptions. HellaSwag is the sole MC case where model×item slightly exceeds item (36.0% vs. 34.3%), though their sum (70.3%) remains in the MC range (68–72%). Among MC benchmarks, ARC-Challenge shows an anomalously high model×prompt interaction (8.0% vs. <2% for MMLU and HellaSwag), suggesting that science-domain items are differentially sensitive to option formatting across model families.

FF benchmark patterns. Free-form benchmarks differ markedly from MC. On GSM8K, variance distributes across item (16.9%), model×item (14.6%), model (10.7%), and model×prompt (8.5%); prompt main effect is 5.4%. MATH shows an intermediate pattern: item leads at 24.0% with model×item at 17.3%, but residual variance is substantially larger (43.9%). Item-level regression reveals that FF residual variance is predominantly explained by the Bernoulli variance ceiling $p(1-p)$ ($R^2 = 0.47$–0.86; Appendix G): medium-difficulty items maximize binary outcome variance regardless of facet structure, so the elevated FF residuals reflect measurement resolution rather than unmodeled substantive sources.

Difficulty dispersion as variance predictor. Difficulty dispersion ($\text{std}(p_i)$) strongly predicts item variance share ($\rho = 0.82$, $n = 5$ benchmarks; Appendix E; reasoning-heavy subjects show G_{item} as low as 0.614, Appendix F). Format constrains the noise floor, but difficulty heterogeneity determines variance magnitude within it.

Item-level bootstrap stability. Bootstrap resampling (1,000 item resamples per benchmark) confirms that the MC–FF separation is robust to item selection: 95% CIs for item-related variance (item + model×item) are [65.2, 75.0]% (MC) and [28.0, 44.2]% (FF), with no overlap (minimum gap 21.0 percentage points).

E Cross-Benchmark Difficulty Heterogeneity

Figure 5 plots item difficulty heterogeneity ($\text{std}(p_i)$ across models) against `item_id` variance share for all five benchmarks. MC benchmarks cluster at higher dispersion (0.24–0.31) and higher item variance (34–38%), while FF benchmarks occupy the lower-left region (0.22–0.23, 17–24%). The cross-benchmark Spearman $\rho = 0.82$ ($p = 0.089$, $n = 5$) is consistent in direction with the within-MMLU correlation ($\rho = 0.806$, $p = 0.005$, $n = 10$ subjects); these are independent analyses at different granularities (benchmarks vs. subjects within MMLU), providing multi-scale support for the item difficulty heterogeneity mechanism discussed in §4.3.

F MMLU Subject-Level Variance Decomposition

Table 10 reports per-subject variance decomposition for 10 MMLU subjects (20 items each, 8 models, 2,304 conditions per item). G_{item} ranges from 0.614 (`high_school_mathematics`)

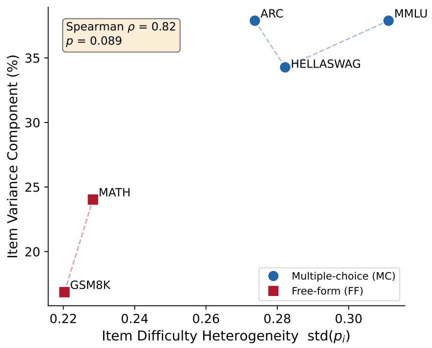

Figure 5: Item difficulty heterogeneity (std(p_i), computed over available models per benchmark, 6–8 models) versus `item_id` variance share (as reported in Table 3) across five benchmarks. MC benchmarks (filled) exhibit higher difficulty dispersion and item variance than FF benchmarks (open). Spearman $\rho = 0.82$, $p = 0.089$.

to 0.930 (`international_law`), with mean 0.843. Knowledge-intensive subjects (`international_law`, `clinical_knowledge`, `world_religions`) show `item_id` > 40%, while reasoning-heavy subjects (`high_school_mathematics`, `abstract_algebra`) show model×item > `item_id`, consistent with models differing more in reasoning strategies than in factual recall. The `high_school_mathematics` `item_id` of 10.4% referenced in §4.3 is the lowest across all subjects, below even GSM8K (16.9%).

Table 10: Subject-level variance decomposition within MMLU (20 items per subject, 8 models, 2,304 conditions per item). Sorted by G_{item} descending.

Subject	Acc.	item%	mod.×it.%	G_{item}
international_law	.717	44.4	24.1	.930
clinical_knowledge	.581	47.5	30.0	.923
world_religions	.826	42.2	27.1	.910
computer_security	.740	37.0	24.9	.908
college_physics	.436	31.6	30.1	.876
philosophy	.782	33.0	36.1	.868
abstract_algebra	.457	24.1	32.8	.822
us_foreign_policy	.864	20.4	36.9	.802
professional_medicine	.671	19.9	41.7	.778
high_school_math	.385	10.4	41.9	.614
Mean	—	31.1	32.6	.843

G FF Residual Variance Analysis

To investigate whether the elevated residual variance in free-form benchmarks (GSM8K 39.9%, MATH 43.9%) reflects unmodeled substantive sources or measurement properties of binary scoring, we regress item-level residual variance on item characteristics across all five benchmarks (Table 11).

The dominant predictor is the quadratic difficulty effect ($p < 10^{-5}$ on all benchmarks): items near $p = 0.5$ show maximal residual variance, while items near the floor or ceiling show minimal residual. This inverted-U pattern directly mirrors the Bernoulli variance ceiling $p(1-p)$. Item content features (number of reasoning steps, answer length) are not significant predictors for GSM8K ($p > 0.38$), confirming that the difficulty–residual relationship is statistical rather than content-driven.

FF benchmarks show substantially higher R^2 (mean 0.66) than MC benchmarks (mean 0.25), consistent with FF items spanning a wider difficulty range. The MC–FF contrast in residual magnitude is therefore a mathematical consequence of the difficulty distribution under binary scoring, not evidence of additional unmodeled variance sources in free-form evaluation.

Table 11: Item-level residual variance regression. Difficulty^2 = quadratic (inverted-U) term reflecting the Bernoulli $p(1-p)$ ceiling. GSM8K additionally tests n_steps and answer_length (both $p > 0.38$).

Benchmark	R^2 (full)	R^2 (diff. only)	Diff.^{2} p
Multiple-choice			
MMLU	0.377	0.347	<0.001
ARC	0.221	0.204	<0.001
HellaSwag	0.156	0.140	<0.001
Free-form			
GSM8K	0.468	0.449	<0.001
MATH	0.856	0.845	<0.001

H Model and Prompt Details

Models. Table 12 lists the eight models with their HuggingFace identifiers.

Table 12: Models used in all experiments.

Short name	HuggingFace identifier	Params
Llama-3.1-8B	`meta-llama/Llama-3.1-8B-Instruct`	8B
Gemma-2-9B	`google/gemma-2-9b-it`	9B
Mistral-7B-v0.3	`mistralai/Mistral-7B-Instruct-v0.3`	7B
Qwen3-8B	`Qwen/Qwen3-8B`	8B
Yi-1.5-9B	`01-ai/Yi-1.5-9B-Chat`	9B
OLMo-2-7B	`allenai/OLMo-2-1124-7B-Instruct`	7B
InternLM2.5-7B	`internlm/internlm2_5-7b-chat`	7B
DeepSeek-7B	`deepseek-ai/deepseek-llm-7b-chat`	7B

Prompt templates. Our six prompt templates follow a surface perturbation design inspired by BrittleBench (Romanou et al., 2026). For multiple-choice benchmarks, templates vary along three dimensions: option layout (lettered vs. numbered), instruction wording (direct vs. contextual), and answer prompt cue. For free-form benchmarks, templates vary chain-of-thought elicitation strategy. All six templates are semantically equivalent—they request the same task with different surface forms—to isolate formatting effects from content effects. ARC-Challenge and HellaSwag use the same six surface patterns as MMLU with domain-appropriate system instructions; MATH mirrors the GSM8K pattern with `\boxed{}` answer formatting.

MMLU templates (representative MC format):

1. `The following are multiple choice questions (with answers) about {subject}.\n\n{question}\nA. ... D. ...\nAnswer:`
2. `Answer the following {subject} questions by selecting the correct option.\nQuestion: {question}\n(A) ... (B) ... (C) ... (D) ...\nCorrect answer:`
3. `Below are {subject} multiple-choice problems. Pick the right letter.\nQ: {question}\n A) ... B) ... C) ... D) ...\nA:`
4. `Answer the following {subject} question by selecting A, B, C, or D.\n{question}\n- A. ... - B. ... - C. ... - D. ...\nAnswer:`
5. `Below is a {subject} exam question with four possible answers. Choose the correct one.\n{question}\nOption A: ... Option B: ... Option C: ... Option D: ...\nCorrect:`
6. `Question ({subject}):\n{question}\nChoices: (A) ... (B) ... (C) ... (D) ...\nYour answer:`

GSM8K templates (representative FF format):

1. `Solve the following math problem step by step.\nQ: {question}\nA: Let's think step by step.`
2. `Answer the following math word problem. Show your work.\nQuestion: {question}\nSolution:`
3. `Solve each problem below. Write your reasoning, then give the final answer after ####.\nProblem: {question}\nReasoning:`
4. `You are a math tutor. Solve the problem and show your steps.\n{question}\nSteps:`
5. `Work through this math question carefully.\nQ: {question}\nWork:`
6. `Math problem:\n{question}\nLet's solve this step by step.`

I Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform related-work section.